

SAHAY-AI

Literature Foundations, Experimental Shadow Classifier and Evaluation Report

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What this report is, and what it is not

- Every model number here is **development evidence**. It is either synthetic (fictional records from a template generator) or *contaminated* regression on fixtures whose outcomes have already been published. None is an official, independent, blind, locked or clinical result. **The locked evaluation set holds 0 samples.**
- The trained model, `experimental_shadow_classifier`, has the status `rejected_for_product_integration`. It runs only in an operator-run local CLI demonstration, beside the deterministic pipeline, which remains the authority.
- SAHAY-AI does not diagnose PTSD, depression or any other condition. It claims no clinical validity, calibration or production readiness.

Abstract

SAHAY-AI is a bounded, human-in-the-loop intake and triage prototype for India’s National Helpline Against Atrocities (NHAA, 14566). This report does three things. First, it reviews the eight references that ground the project: speech-based PTSD and emotion detection, the helpline’s operational mandate, India’s data-protection law, a multilingual encoder for Indian languages, open speech recognition, and a Hindi counselling-dialogue emotion corpus. Second, it documents the one machine-learning model the team trained: an eight-label multilingual *shadow* classifier built on MuRIL. The encoder was domain-adapted with masked-language modelling on 512,605 external text windows, then supervised on fictional, template-generated SAHAY records. Third, it reports every evaluation that was performed, keeping each evidence class separate.

Domain adaptation reduced held-out masked-language perplexity from 11.95 to 6.17. The Task 7 checkpoint reached micro-F1 0.78 on its synthetic development test, but crisis recall on validation was only 0.23. On the exposed regression fixtures it lagged the deterministic lexicon pipeline: micro-F1 0.67 against 0.91 on dev, over the seven shared labels. A targeted hardening round (Task 7B) raised crisis recall on a frozen synthetic holdout to 1.00 in English, Hindi and Hinglish. It still missed the pre-registered macro-F1 gate (0.675 against 0.70), cut continuing-threat recall to 0.11, and could not be evaluated on three labels that had no positive holdout support. It also did worse than Task 7 on the exposed fixtures. Both checkpoints therefore remain rejected for product integration.

The deterministic pipeline remains the authority. After safety hardening it misses 1 of 17 critical events on the dev fixtures and 1 of 14 on the (now exposed) candidate fixtures, and it passes all 39 exposed guardrail red-team cases. The report closes with limitations and a concrete path to an independent evaluation.

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1 Introduction

1.1 Operational context

The Ministry of Social Justice and Empowerment (MoSJE) runs the National Helpline Against Atrocities (NHAA) on the toll-free number 14566 [3]. The helpline supports enforcement of the Protection of Civil Rights (PCR) Act, 1955 and the Scheduled Castes and Scheduled Tribes (Prevention of Atrocities) Act, 1989. It is available around the clock across the country, in Hindi, English and the regional languages of the States and UTs.

Problem Statement 26093 (MoSJE) asks for an “*AI-enabled Real-Time Stress and Trauma Assessment Module*” that assesses stress, trauma, fear, anxiety and vulnerability across NHAA channels. It names voice and text analysis, a vulnerability index with Low/Moderate/High/Critical bands, detection of suicidal ideation and intimidation, recommended support pathways, and multilingual operation under privacy and consent constraints.

1.2 The SAHAY-AI design stance

SAHAY-AI treats that brief as a *bounded intake instrument*, not a counsellor or a diagnostic tool:

- A deterministic 12-state dialogue machine chooses every approved intent. A language model may only phrase that intent, and a validator checks every sentence before it is spoken.
- A synchronous crisis pre-check runs before dialogue policy on every utterance. A match forces the fixed crisis script, Critical priority and a human takeover.
- A deterministic Stress Vulnerability Index (SVI) combines nine evidence-linked dimensions, D1–D9. It abstains (`needs_human`) when confidence is low.
- A human executive decides every action. The victim never sees any score, band, alert or confidence.

These are the project’s non-negotiable safety invariants. The model in this report is judged against them: it may only *add evidence beside* the deterministic pipeline, never replace it.

1.3 Why train a model at all?

The deterministic detectors are precise, but they are brittle to paraphrase, indirect wording and code-switching. Before hardening, they missed 4 of 14 critical events on the candidate fixtures (§6.6). A learned multilingual encoder is the natural complement, but only if it can be *shown* to help. The team therefore trained it strictly in shadow mode, under a pre-registered selection and promotion protocol. The protocol is authorised by decision EXT-119 and its Invariant 8 clarification.

1.4 Evidence classes

Every number in this report belongs to exactly one of four evidence classes. They are never pooled (Table 1).

1.5 Structure

Section 2 reviews the eight references. Section 3 summarises the system. Section 4 describes the data, and Section 5 the model and its training. Section 6 reports all evaluations. Sections 7–9 discuss findings, limitations and next steps. The appendices hold the provenance hashes and the label definitions.

Table 1: Evidence classes used throughout the report.

Class	What it is	What it can and cannot support
A External auxiliary	Masked-language perplexity and source-namespace auxiliary metrics on external datasets	Evidence of language modelling only. <i>Never</i> a SAHAY safety metric: source labels such as subreddit, stress, emotion, sentiment or hate speech are not SAHAY labels.
B Synthetic development	Metrics on fictional records from the same template generator (validation, <code>synthetic_development_test</code> , the Task 7B frozen holdout)	Shows the model learned the generator’s patterns. Says <i>nothing</i> about real speech or text.
C Contaminated regression	The published dev, candidate and red-team fixtures, run once after selection	A regression check next to the deterministic pipeline. Never used for training, early stopping, thresholds or selection.
D Official / independent	The locked set and the human-authored blind corpus	Empty: the locked count is 0 and nothing has been frozen, so no official number exists.

2 Literature and reference review

The eight references fall into five themes. For each one we summarise the verified facts, say how it bears on SAHAY-AI, and appraise it critically. All details were checked against the publisher, abstract or repository page on 13 September 2026.

2.1 Speech markers of PTSD and emotion

Hu et al. (2024) [1]. *Speech-based recognition and estimating severity of PTSD using machine learning*, Journal of Affective Disorders 362:859–868. The authors recorded speech from **76 patients with PTSD and 60 controls**. They extracted acoustic features with the openSMILE toolkit and selected features with a random forest. Classifiers on an 18-feature set gave strong binary results: the random-forest model reached **accuracy 0.975** and **AUC 1.0**. A regression model predicting PCL-5 severity scores was statistically significant, but weak (**MSE 0.90**, **MAE 0.76**, $R^2 = 0.10$, $p < 0.001$). The authors themselves note that feature selection may inflate the results, that the underlying mechanisms are unexplained, and that the regression needs validation on larger datasets.

Islam and ElSayed (2024) [2]. *Speech-Based Emotion Recognition and PTSD Detection through Machine and Deep Learning*, IJCERT 11(3):46–53. The study combined the acted RAVDESS emotional-speech corpus with PTSD-specific recordings, after noise reduction. It extracted acoustic and temporal features and compared SVMs and random forests against CNNs and LSTMs. Deep models reached **up to 91% accuracy for emotion recognition** and **89% for PTSD detection**, clearly ahead of the classical models. The authors flag limited generalisability across populations and obstacles to real-world deployment.

Critical appraisal. Both studies show that acoustic signals carry information about trauma-related states, but three cautions apply before carrying their numbers over to a helpline:

- **Small or acted samples.** With 136 speakers, an AUC of exactly 1.0 is a warning sign of selection optimism, which the authors acknowledge. RAVDESS is acted, scripted English speech; acted emotion is not the spontaneous distress of a first-contact call in Hindi or Hinglish.
- **Weak severity signal.** $R^2 = 0.10$ means acoustic features explain about a tenth of the variance in symptom severity. That argues *against* deriving a severity score from voice.

- **Provenance.** The source of the PTSD recordings in [2] is not described in its abstract, so the result cannot be reproduced from the paper’s summary alone.

Implications for SAHAY-AI. The two studies motivate SVI dimension D4, *acute distress (acoustic and emotional)*, weight 0.12. SAHAY-AI deliberately treats it as a distress *indicator*, never as a diagnosis. The output validator rejects any assistant sentence that names PTSD, trauma, depression or a disorder as fact. D4 is also currently **structurally unavailable**: no approved, usable acoustic corpus exists. The local CREMA-D copy consists of Git LFS pointers, and RAVDESS (EXT-003) and Common Voice Hindi (EXT-002) are still only proposed. The SVI therefore renormalises over the remaining 0.88 of weight on text sessions. The high headline accuracies in [1, 2] are also why this report insists on denominators, undefined-metric handling and separate evidence classes.

2.2 Operational mandate: NHAA 14566

The MoSJE press release of December 2021 [3] gives these facts about the helpline:

- toll-free number **14566**, operating round the clock nationwide;
- reachable by voice call or VOIP from any mobile or landline operator, and through a web-based self-service portal;
- Hindi, English and the regional languages of the States and UTs;
- a **docket number for every complaint**, with online status tracking;
- the goal of making sure the victim-related provisions of the PoA Act, 1989 and the PCR Act, 1955 are complied with, from complaint registration and FIR through to investigation and charge-sheeting within statutory timelines;
- awareness-building about those legal provisions.

These features map directly onto SAHAY-AI. The multilingual mandate drives the Hindi, English and Hinglish MVP. Docketing and tracking become the status-only *victim timeline*. The statutory-compliance goal drives the legal-urgency dimension (D8) and the escalation packet. SAHAY-AI does not integrate with the live helpline: that needs departmental approval, an authorised operator and DLT registration.

2.3 Legal framework: DPDP Act 2023 and Rules 2025

The Digital Personal Data Protection Act, 2023 (Act 22 of 2023, assented in August 2023) is India’s general data-protection law [4]. MeitY notified the implementing *DPDP Rules, 2025* in mid-November 2025; sources give 13 or 14 November. Commencement is phased over 18 months:

- the Data Protection Board provisions took effect immediately;
- the consent-manager provisions follow after 12 months (around November 2026);
- most substantive obligations follow after 18 months (around May 2027).

Those substantive obligations cover notice and consent, reasonable security safeguards, breach intimation to the Board and to affected Data Principals, verifiable parental consent for children’s data, retention limits and erasure, and extra duties for Significant Data Fiduciaries. Penalties reach INR 250 crore.

Separately, the licensing and privacy status of the external training datasets is **unresolved** (§4). EXT-119 records a project-owner risk acceptance for private, local research only. It is not a lawful-basis determination.

2.4 Multilingual text representation: MuRIL

MuRIL (Khanuja et al., Google Research India, arXiv:2103.10730) [5] is a BERT-base encoder pre-trained only on Indian-language text:

- **17 languages:** 16 Indian languages plus English;

Table 2: How DPDP principles map to SAHAY-AI controls (design alignment only; not legal advice).

DPDP principle	SAHAY-AI control (MVP)
Notice and informed consent	Consent gate with AI disclosure before any capture; scoring suppressed if consent is declined; “Talk to a person” always visible.
Purpose limitation, minimisation	Victim clients never receive the SVI, bands, dimensions, alerts or confidence (enforced server-side and tested); queue previews minimised.
Security safeguards	All speech and NLP runs locally; audio never leaves the deployment; role-filtered event fan-out; no keys in the repository.
Retention and erasure	Audio kept for less time than derived case data; the training artefacts have a documented keep, quarantine and delete policy.
Children’s data	The external training datasets carry a <code>minors_possible</code> flag; they stay quarantined research artefacts and never reach the product.
Breach intimation, Board	<i>Not implemented.</i> A production deployment would need a breach process and a legal review.

- **236M parameters**, a **197,285-token vocabulary**, sequences of up to 512 tokens, and about 16B unique training tokens;
- two objectives: masked-language modelling (MLM) on monolingual text and translation-language modelling (TLM) on parallel data;
- data from Common Crawl (OSCAR) and Wikipedia, translated pairs (PMINDIA and in-house translations) and transliterated pairs (Dakshina, indic-trans);
- low-resource languages upsampled with exponent $\alpha = 0.3$.

On XTREME-style tasks MuRIL beat mBERT with an average of **68.6 vs 59.1** on native-script test sets and **48.9 vs 21.1** on transliterated test sets.

Why SAHAY-AI uses it. Helpline text mixes Devanagari Hindi, English and *romanised* Hinglish. MuRIL’s large advantage on transliterated input is exactly the property we need. Our own runtime qualification (§6.7) found that the pinned MuRIL tokeniser uses about 23% fewer tokens per character than XLM-R on Hindi (0.246 vs 0.319) and about 12% fewer on Hinglish (0.259 vs 0.294), with a 0% unknown-token rate. Our Stage A sampler reuses MuRIL’s upsampling exponent $\alpha = 0.3$, applied across source datasets instead of languages (§5.2).

2.5 Speech recognition: IndicConformer and faster-whisper

IndicConformer [6]. AI4Bharat’s IndicConformer is a suite of open ASR models covering all **22 official Indian languages**. The multilingual model has **600M parameters** and uses hybrid RNN-Transducer models built on the Conformer architecture. It is released under the MIT licence through Hugging Face and a direct object store. In SAHAY-AI it is the *roadmap* path to the 22 scheduled languages. It has no external-dependency approval, has **not been downloaded**, and no WER has been measured with it.

faster-whisper [7]. SYSTRAN’s faster-whisper reimplements OpenAI Whisper on the CTranslate2 inference engine. It is up to $4\times$ faster than the reference implementation at the same accuracy, uses less memory, supports 8-bit quantisation on CPU and GPU, and integrates Silero VAD (MIT licence). Its README benchmark for the *small* model on CPU shows why int8 matters for a laptop fallback:

- fp32, beam 5: 2 min 37 s and 2,257 MB RAM;
- int8, beam 5: 1 min 42 s and 1,477 MB, about 35% less time and memory;
- int8 with batch 8: 51 s and 3,608 MB.

SAHAY-AI runs Whisper Small through faster-whisper 1.2.1 and CTranslate2 4.8.2. The official checkpoint was converted locally to CTranslate2 fp16 on the GPU, with a CPU int8 fallback labelled

degraded. The *only* application entry point is Silero VAD gating (§6.7).

2.6 Hindi emotion in counselling dialogue: EmoInHindi

EmoInHindi (Singh, Priya, Firdaus, Ekbal and Bhattacharyya, IIT Patna; LREC 2022, pp. 5829–5837) [8] contains **1,814 dialogues** and **44,247 utterances**. They were created in a Wizard-of-Oz manner *for mental-health and legal counselling of crime victims*. Each utterance carries one or more of 16 emotion labels (including neutral) and an intensity. The paper also proposes contextual baselines for emotion and intensity recognition.

Relevance and use. EmoInHindi is the public resource closest to SAHAY-AI’s domain: Hindi, crime-victim, counselling-style dialogue. Under EXT-119 it was used only for Stage A masked-language modelling (10,815 coverage windows, 14.6% of continuation draws) and for an *isolated* Stage B auxiliary head (§5.3). Its emotion labels are never mapped to D4 (which is acoustic), the SVI, a band or crisis. The publisher permits research use only and does not address redistribution, so the dataset stays `licence_pending`. One local observation is unreconciled: the processed local copy exposes 17 label names (Stage B vocabulary size 17), while the paper states 16.

2.7 Synthesis

Table 3: The eight references: verified findings, role in SAHAY-AI, and current status.

#	Reference	Key verified finding	Role in SAHAY-AI	Status in project
1	Hu et al. 2024 [1]	76 PTSD / 60 controls; openSMILE features + RF; acc. 0.975, AUC 1.0; PCL-5 regression R^2 0.10	Motivates the D4 acoustic channel; argues against voice-derived severity	Not reproduced; D4 unavailable
2	Islam & ElSayed 2024 [2]	RAVDESS + PTSD recordings; deep models up to 91% (emotion), 89% (PTSD)	Speech emotion as an auxiliary signal only; acted-speech caveat	Not reproduced; RAVDESS not approved
3	MoSJE / PIB [3]	NHAA 14566: 24×7, toll-free, multilingual; docket and tracking; PCR / PoA Acts	Deployment context, timeline, D8 legal urgency	Context only; no live integration
4	MeitY DPDP [4]	Consent, notice, safeguards, breach, children; Rules 2025 phased to 2027	Privacy-by-design controls (Table 2)	Alignment documented; legal review pending
5	Khanuja et al. 2021 [5]	17 languages, 236M params, MLM+TLM; translit. 48.9 vs mBERT 21.1	Base encoder of the trained shadow classifier	Used (rev. afd9f36c , Apache-2.0)
6	AI4Bharat [6]	IndicConformer 600M, 22 languages, Conformer hybrid RNNT, MIT	Roadmap ASR for 22 languages	Not downloaded; not benchmarked
7	SYSTRAN [7]	CTranslate2 Whisper, up to 4× faster, int8, Silero VAD	ASR runtime for Whisper Small	Used (v1.2.1, VAD-gated)
8	Singh et al. 2022 [8]	1,814 WoZ dialogues, 44,247 utterances, 16 labels + intensity; crime-victim counselling	Hindi domain text for Stage A / B	Used under EXT-119; licence pending

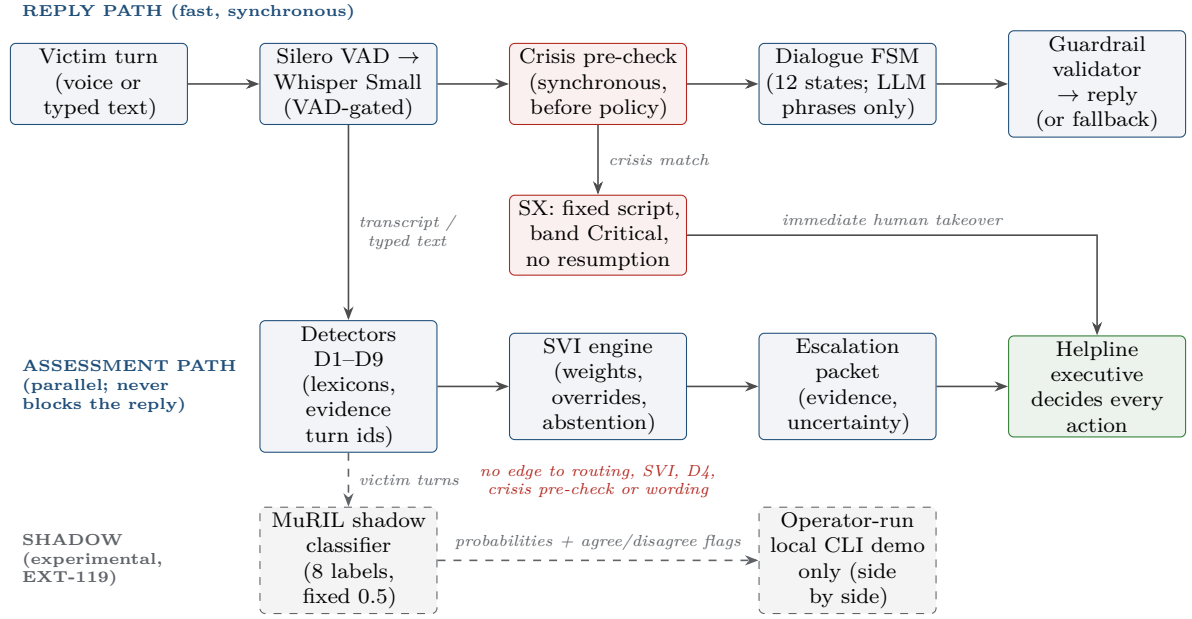


Figure 1: SAHAY-AI architecture and where the trained model sits. Solid arrows are authoritative paths; dashed arrows are shadow-only. The deterministic pipeline runs unchanged with every model absent.

3 System context

Figure 1 places the trained model inside the SAHAY-AI architecture. It has two paths. The *reply path* is synchronous and protected by the crisis pre-check. The *assessment path* runs in parallel and never blocks a reply. The shadow classifier sits entirely outside both decision paths.

The Stress Vulnerability Index. The SVI is a deterministic weighted sum, $SVI = \sum_i w_i s_i$, with each dimension score $s_i \in [0, 100]$ (Table 4). The bands are Low (0–29), Moderate (30–54), High (55–74) and Critical (75–100). *Hard overrides* sit outside the sum: a confirmed D1 (immediate danger) or D2 (crisis) forces Critical, and confidence below 0.45, poor audio or low language confidence yields **needs_human** with *no score*. On typed channels D4 is structurally unavailable, so the weights renormalise over the remaining 0.88.

Table 4: SVI dimensions and provisional weights (pending expert calibration).

Dim	Meaning	Weight	Dim	Meaning	Weight
D1	Immediate safety threat	0.22	D6	Isolation, boycott, displacement	0.08
D2	Crisis / self-harm language	0.18	D7	Medical urgency	0.08
D3	Fear, intimidation, threats	0.13	D8	Legal urgency	0.06
D4	Acute distress (acoustic, emotional)	0.12	D9	Communication safety	0.05
D5	Trauma-associated indicators	0.08			

The eight detector categories. Both the deterministic pipeline and the shadow model are scored on the same schema. It has eight multi-label categories: crisis / self-harm, immediate danger, continuing threat, medical urgency, isolation / boycott / displacement, legal urgency, communication-safety coercion, and explicit human request (definitions in Appendix B). The deterministic pipeline has no text detector for *explicit human request*, because the app sends that request from a button. Deterministic metrics therefore cover seven categories.

4 Data

4.1 External corpora (Stage A and B only)

EXT-119 authorised local, offline training on every usable, privacy-processed text record of five external datasets (Table 5). All five remain `licence_pending`, and every output is a `quarantined_research_artifact`. The builder counts blank rows, unreadable bytes and LFS pointers but never invents them. Text is split into windows of at most 480 characters on word or turn boundaries, so each fits a 128-token input.

Table 5: External datasets used for encoder adaptation (Stage A) and isolated auxiliary heads (Stage B). *Dialogues. †15,000 labelled records plus 3,199 unlabelled ones used for MLM only.

Dataset	Language / script	Records	Windows	Dup. recs	Stage B auxiliary task (source namespace)
Reddit Suicide Detection	English	232,072	466,055	519	Subreddit membership (not a clinical judgement)
Dreaddit	English	715	968	0	Annotated stress (not crisis)
EmoInHindi [8]	Hindi (mostly mixed script)	1,814*	10,861	4	Multi-label emotion (not severity)
Hinglish hate-speech (local derivative)	Romanised	18,199†	20,151	2,912	Hate-speech label (not threat or danger)
Hinglish sentiment	Romanised Hinglish	14,569	14,570	684	None: classes 0/1/2 are opaque
CREMA-D	—	0	—	—	Unavailable: all 22,326 media files are Git LFS pointers
Total		267,369	512,605		263,904 duplicate families

Governance.

- Every window is weighted by $1/\text{family size}$, so near-duplicates do not dominate.
- About 0.5% of families are held out as `mlm_validation` and about 10% form an auxiliary test bucket. No family crosses partitions.
- The privacy pass redacts URLs, e-mail addresses, phone numbers, long numbers and social handles. Examples: 11,556 Reddit records were redacted, and 955 EmoInHindi records. Rule-based redaction *cannot* catch names or contextual identifiers.
- A **source-label firewall** refuses every automatic mapping from a source label to a SAHAY concept. For example: `stress` $\nrightarrow$ `crisis`, `subreddit of origin` $\nrightarrow$ `routing`, `sentiment` $\nrightarrow$ `SVI`, `emotion` $\nrightarrow$ `D4`, `hate speech` $\nrightarrow$ `continuing threat`, and “non-suicide” $\nrightarrow$ `safe`.

4.2 Fictional SAHAY corpus (Task 7, Stage C)

The SAHAY head was supervised *only* on private fictional records. They come from a committed bank of **110 templates** in English, Devanagari Hindi and romanised Hinglish:

- **Size:** 6,605 records, split *by template* into 5,005 train, 757 validation and 843 `synthetic_development_test`.
- **Coverage:** single- and multi-turn records, multi-label cases, safe and low-distress controls, expected-abstention cases and human-help requests. Also negation, quotation, attribution, past events, indirect and conditional wording, misspellings, Unicode variants and code-switching.
- **Contamination screen:** every rendered variant was checked against the exposed fixtures. Hits (39 shared distinctive phrases, 8 high token overlaps) blocked **5 template families** (115 records). Blocked families were left out, never reworded to evade the check.

4.3 Targeted hardening corpus (Task 7B, 7b-v1)

An ID-only error analysis of the Task 7 checkpoint found crisis misses concentrated in Hindi and Hinglish, especially in conditional (“temporal ambiguity”), indirect and “disappear” wording. It also found coercion confused with ordinary disagreement, and legal help confused with legal urgency. A targeted corpus answered this:

- **Families:** 105 fictional cores with explicit contrast families. Direct framings keep their label; quoted, reported, past and negated framings flip it.
- **Size:** 9,924 records: 35% English, 28% Hindi, 38% Hinglish.
- **Focus areas:** crisis 34%, coercion 21%, legal 19%, multi-label 13%, other 14%.
- **Freeze:** the corpus was frozen and hashed *before* training (2026-09-12 11:01:49 IST).
- **Review packet:** 2,266 items across 718 families were prepared, but **0 human reviews are complete**.

Table 6: Task 7B split sizes and positive label support. Three labels have *no* positives in either validation or holdout.

Split	Records	Families	Crisis	Imm. dgr	Cont. thr	Medical	Isolation	Legal	Coercion	Human req.
Train	7,492	634	2,108	268	453	254	246	1,200	1,441	370
Validation	1,242	42	318	0	243	0	0	212	236	105
Holdout	1,190	42	298	0	132	0	0	209	216	118

4.4 Deterministic evaluation corpora

Label schema 1.0.0 defines these corpora, all of them fictional:

- **dev:** 57 samples (13 Hindi, 28 English, 16 Hinglish); tuning allowed;
- **candidate:** 48 samples (12 Hindi, 22 English, 14 Hinglish); never tuned on, but every outcome has been published, so the split is exposed;
- **locked:** **0 samples**, because no fixture has the required real reviewer approvals;
- **red-team:** 39 guardrail cases, plus 75 near-miss cases and 29 urgency-leakage cases written during hardening, plus 8 red-team *victim-input* cases.

A blind-evaluation harness exists (`ml/eval/blind/`). Its `run` command refuses with exit code 12 because no corpus has been frozen.

5 Model and training

5.1 Architecture

`experimental_shadow_classifier` is built from four parts:

- **Encoder:** MuRIL-base, pinned at revision `afd9f36c` [5]: 12 layers, hidden size 768, 12 attention heads, feed-forward size 3,072, vocabulary 197,285. It was domain-adapted in Stage A.
- **Pooling:** attention-masked mean pooling, then dropout 0.1.
- **Head:** one linear layer with **8 independent logits** (sigmoid, BCE-with-logits), no routing, SVI, D4, diagnosis or generative head.
- **Input and threshold:** victim turns only, joined in order, truncated to 128 tokens. The decision threshold is fixed at an **uncalibrated 0.5** and never tuned.

Training used PyTorch 2.11.0+cu128 and Transformers 5.17.0 on a single laptop RTX 4060 (8 GB), fully offline with sockets blocked (0 network attempts recorded).

5.2 Stage A: domain-adaptive masked-language modelling

Stage A continues MuRIL’s MLM objective on the external corpus in two phases:

1. a seeded *coverage pass* of 277,185 training windows, so every usable record contributes at least its first window;
2. a *balanced continuation* of 40,000 windows, each dataset drawn with probability proportional to its window count raised to the power 0.3. MuRIL uses the same exponent for language upsampling [5]; here it is applied across datasets.

The continuation cut the suicide corpus to 45.0% of draws. Hinglish hate speech took 17.5%, Hinglish sentiment 15.9%, EmoInHindi 14.6% and Dreddit 7.0%; Dreddit’s small pool was cycled through four passes.

Table 7: Stage A configuration and results (evidence class **A**).

Setting	Value
Masking	Dynamic 15% (80% [MASK], 10% random, 10% unchanged); loss on masked tokens, weighted by duplicate family
Optimiser, schedule	AdamW (fused), learning rate 5×10^{-5} , weight decay 0.01, 6% warmup then linear decay, gradient clip 1.0
Batch, length, precision, seed	Micro-batch 32, 128 tokens, bf16 autocast with fp32 master weights, seed 20260912
Optimiser steps, wall time	9,913 (8,663 coverage + 1,250 continuation); 3,606 s (≈ 60 min) at ≈ 91 windows/s
Peak GPU memory	5,923 MB allocated, 9,134 MB reserved (above the 8 GB of dedicated VRAM, so it spilled into shared memory); max 57 °C
Validation loss / perplexity (2,732 windows, 33,197 masked tokens)	before: 2.480 / 11.945 ; after coverage: 1.854 / 6.385; after continuation: 1.819 / 6.165 (−48.4% perplexity)

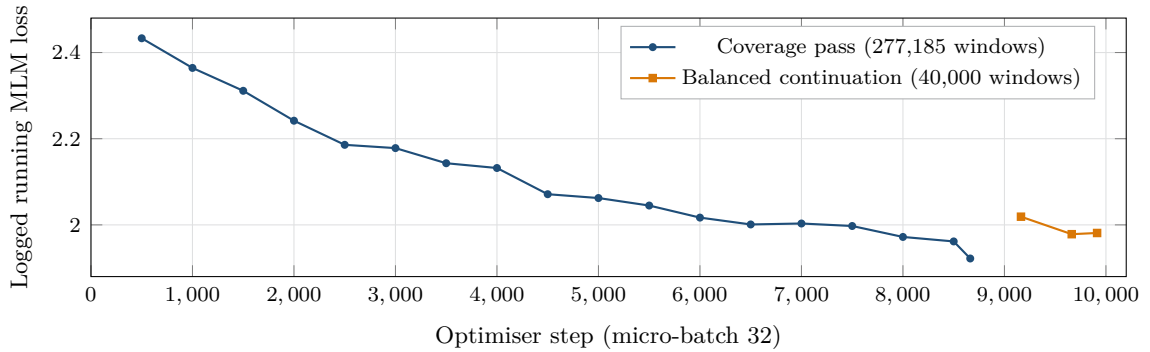


Figure 2: Stage A training loss from the run console. The small step up at the start of the continuation most likely reflects the shift in data mix toward the smaller Hinglish, Hindi and Dreddit sources.

A 2,048-window smoke run was done first (perplexity 14.03 $\rightarrow$ 10.37 on its smaller validation set). It is not used.

5.3 Stage B: isolated auxiliary heads

Stage B trained one head per source dataset on the Stage A encoder. Labels were kept in their source namespace (`source:<dataset>:<value>`), with at most 12,000 training and 3,000 test windows per source. It is a diagnostic of what the adapted encoder represents. Its checkpoint **never** initialised or touched the SAHAY head. Hinglish sentiment was excluded because its classes are opaque.

On EmoInHindi the head reaches F1 0.88 on *anticipation* and 0.83 on *joy*. It **never predicts** 7 of the 17 labels: annoyed, apprehensive, confused, *fear*, guilty, impressed and surprised, all with F1 0.00. Fear (210 test positives) is exactly the emotion a helpline cares about, which illustrates why emotion labels are not a usable proxy for distress.

Table 8: Stage B source-namespace results (evidence class **A**; not SAHAY safety metrics).

Source	Task	Train win.	Test win.	Micro F1	Macro F1	Exact match
Reddit Suicide Detection	2-class	12,000	3,000	0.900	0.890	0.900
Hinglish hate speech	2-class	12,000	1,652	0.827	0.824	0.827
Dreaddit	2-class	860	108	0.657	0.644	0.657
EmoInHindi	multi-label (17)	9,783	1,078	0.667	0.355	0.049

5.4 Stage C: the SAHAY shadow head (Task 7)

Stage C starts from the Stage A encoder and trains the eight-label head on the fictional corpus. Settings:

- AdamW (fused); encoder learning rate 3×10^{-5} , head learning rate 10^{-3} ;
- linear schedule with 6% warmup; micro-batch 32; bf16; gradient clip 1.0;
- BCE-with-logits loss; seeds 13, 42 and 97.

Two runs are disclosed:

- **Run 1:** at most 6 epochs, patience 2. It hit the epoch cap while validation was still improving, so early stopping never acted.
- **Run 2:** at most 20 epochs, patience 3, everything else identical. It is the one change that lets the declared early-stopping rule actually decide.

The rule, fixed before training, ranks candidates on *validation only*:

1. higher validation macro F1;
2. higher minimum per-label recall;
3. lower validation loss;
4. lower seed, then earlier run.

Table 9: Stage C candidates (evidence class **B**). Selection used validation columns only; synthetic-test columns are shown for transparency. The selected row is shaded.

Run	Seed	Epoch (of run)	Validation (757)				Synthetic dev. test (843)		
			Macro F1	Min. label R	Crisis R	Loss	Macro F1	Micro F1	Crisis R
1	13	6 (6)	0.437	0.013	0.013	0.284	0.532	0.672	0.111
1	42	6 (6)	0.428	0.000	0.091	0.293	0.514	0.659	0.181
1	97	6 (6)	0.321	0.007	0.026	0.353	0.353	0.588	0.042
2	13	15 (18)	0.609	0.197	0.325	0.454	0.806	0.806	0.500
2	42	10 (13)	0.630	0.104	0.260	0.428	0.732	0.763	0.417
2	97	9 (12)	0.645	0.234	0.234	0.373	0.761	0.783	0.458

Three observations follow from Table 9:

- **Loss and F1 disagree.** Run 1 has a *lower* validation loss than Run 2 but far lower macro F1. With imbalanced multi-label BCE, predicting mostly negatives is cheap in loss, so F1-first selection was the right call.
- **Seed variance is large.** Run 2 synthetic-test macro F1 spans 0.73–0.81 across seeds. Differences under about 0.05 should not be read as real.
- **Validation and test disagree.** Seed 13 is best on the synthetic test, but the rule correctly chose from validation only. The two splits also differ in difficulty: validation holds 142 *conditional*-wording records and 153 *indirect* ones, while the test split has no conditional records. That is why validation crisis recall (0.23) is half the test value (0.46).

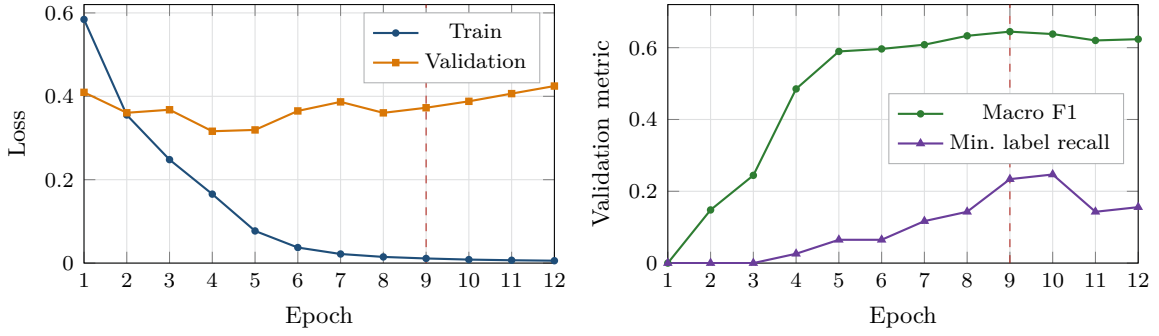


Figure 3: Selected Task 7 run (Run 2, seed 97). The validation loss bottoms out at epoch 4, while train loss keeps falling to near zero: the head memorises the templates. Macro F1 peaks at epoch 9 (dashed), where the rule selected the checkpoint.

5.5 Task 7B: targeted retraining under a pre-registered plan

Task 7B retrained only the Stage C head. It reused the verified Stage A encoder and trained only on the frozen 7b-v1 corpus. The plan was hashed before training (843f3e2c...), and code that no longer matched it refused to run. It fixed:

- **Three configurations:** A, BCE with a class-balanced sampler; B, BCE with `pos_weight` clipped to $[1, 8]$; C, focal loss with $\gamma = 2.0$.
- **Schedule:** seed 13 for all three, then seeds 42 and 97 for the winner.
- **Settings:** micro-batch 16 with accumulation 2, weight decay 0.01, at most 20 epochs, patience 3.
- **Selection rule**, on validation only:
 1. crisis recall;
 2. then the minimum recall over crisis, legal and coercion;
 3. then macro F1, then no-alert specificity, then loss.
- **Promotion gates**, never lowered (Table 13).

Table 10: Task 7B selection ranking (validation only, evidence class **B**). Selected row shaded. [†]Minimum recall over crisis, legal urgency and coercion.

Config (seed)	Epoch	Crisis R	Min. weak R [†]	Macro F1	No-alert spec.	Val. loss
C focal (13)	3	1.000	0.920	0.695	0.914	0.178
A balanced (13)	6	1.000	0.792	0.524	0.909	0.295
C focal (97)	2	0.972	0.674	0.513	0.940	0.239
C focal (42)	5	0.950	0.950	0.674	0.938	0.149
B pos_weight (13)	2	0.918	0.759	0.594	0.957	0.210

6 Evaluation results

6.1 Task 7 checkpoint on synthetic development data (B)

Table 11 gives per-label results for the selected checkpoint. The three weakest labels on validation are crisis, coercion and legal urgency: crisis recall is 0.234 (Wilson 95% CI 0.153–0.340), coercion F1 0.39, legal-urgency F1 0.40. These are the wordings the lexicon pipeline also struggles with.

By language. Validation macro F1 is 0.730 in English, 0.618 in Hindi and **0.503 in Hinglish**. In both Hindi and Hinglish the minimum per-label recall is **0.00**, meaning at least one label is never detected in those languages. On the synthetic test the figures are 0.727 (English), 0.823 (Hindi) and 0.740 (Hinglish). Code-mixed input is the hardest case, which matches the Task 7B error analysis.

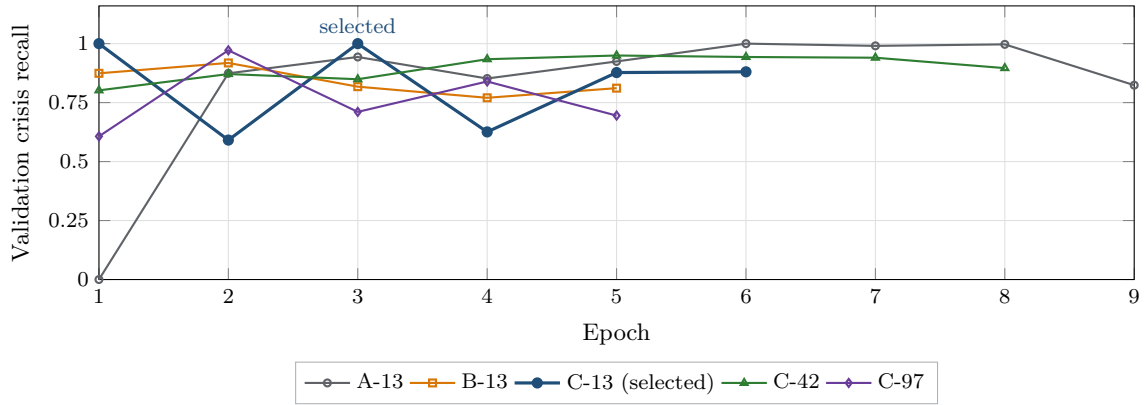


Figure 4: Validation crisis recall by epoch for the five Task 7B runs. The selected run (C-13) swings between 1.00 and 0.59 on alternate epochs, so its selected epoch sits on a spike rather than a plateau. C-42 is the most stable run.

Table 11: Task 7 selected checkpoint (Run 2, seed 97) at the fixed 0.5 threshold. Pos. = positive support.

Label	Validation (757 records)				Synthetic development test (843 records)			
	Pos.	Precision	Recall	F1	Pos.	Precision	Recall	F1
Crisis / self-harm	77	0.439	0.234	0.305	72	0.452	0.458	0.455
Immediate danger	86	0.905	1.000	0.950	119	0.943	0.975	0.959
Continuing threat	218	0.832	0.431	0.568	285	0.720	0.937	0.814
Medical urgency	68	0.850	1.000	0.919	97	0.970	1.000	0.985
Isolation / boycott / displ.	67	0.667	0.657	0.662	87	0.529	0.828	0.646
Legal urgency	71	0.550	0.310	0.396	76	0.604	0.724	0.659
Coercion (comm. safety)	144	0.358	0.431	0.391	151	0.659	0.894	0.758
Explicit human request	84	0.933	1.000	0.966	94	0.759	0.872	0.812
Macro (8 labels)		0.692	0.633	0.645		0.705	0.836	0.761
Micro		0.685	0.587	0.632		0.710	0.874	0.783

6.2 Task 7 on the exposed regression fixtures (C)

The checkpoint was run once, after selection, on the published fixtures, next to the deterministic pipeline (Table 12). One correction applies to the stored summaries. The shadow micro-F1 values in the model card (0.68 dev, 0.70 candidates) include the eighth label, explicit human request, which the deterministic pipeline does not score. Recomputed from the stored per-label counts over the *seven shared labels*, they are 0.673 and 0.683. We use the seven-label values for comparison.

Table 12: Per-label F1 on the exposed fixtures (evidence class **C**; contaminated, not generalisation evidence). “Deterministic” is the pipeline after safety hardening. Best value per row in bold.

Label	Dev fixtures (57)			Candidate fixtures (48)		
	Deterministic	Task 7	Task 7B	Deterministic	Task 7	Task 7B
Crisis / self-harm	0.778	0.714	0.583	0.842	0.889	0.720
Immediate danger	0.933	0.857	0.333	1.000	0.727	0.600
Continuing threat	0.889	0.800	0.621	0.957	0.667	0.667
Medical urgency	1.000	0.800	0.667	0.800	0.750	0.800
Isolation / boycott / displ.	0.889	0.364	0.727	0.667	0.857	0.400
Legal urgency	1.000	0.462	0.500	0.923	0.400	0.571
Coercion (comm. safety)	0.923	0.400	0.500	0.571	0.286	0.462
Explicit human request	n/a	0.750	1.000	n/a	1.000	0.800
Micro F1, 7 shared labels	0.906	0.673	0.564	0.868	0.683	0.622
Micro F1, as stored (shadow incl. human req.)	—	0.678	0.585	—	0.705	0.632
Exact match	0.860	0.632	0.544	0.792	0.667	0.542

The deterministic pipeline wins on 12 of the 14 label–split cells. The Task 7 shadow is ahead only on

candidate crisis (0.889 vs 0.842) and candidate isolation (0.857 vs 0.667). On the 8 red-team *victim-input* cases, the Task 7 crisis firing matched expectation in 6 of 8. It falsely fired on RT-VI-EN-002 and missed the Hinglish crisis case RT-VI-HG-001.

6.3 Task 7B on the frozen synthetic holdout (B, evaluated once)

The selected Task 7B checkpoint was evaluated *once* on the 1,190-record frozen holdout. The old Task 7 checkpoint was evaluated on the same holdout for comparison.

Table 13: Task 7B promotion gates on the frozen holdout. Any failed gate keeps the model rejected; even a full pass gives only `candidate_for_human_review`.

Gate	Required	Task 7B value	Result
Holdout crisis recall	≥ 0.80	1.000 (298/298; Wilson 95% CI 0.987–1.000)	pass
Crisis recall, English / Hindi / Hinglish	≥ 0.70 each	1.000 / 1.000 / 1.000	pass
Legal-urgency recall	≥ 0.70	0.727 (152/209)	pass
Coercion recall	≥ 0.70	1.000 (216/216)	pass
Macro F1 (5 labels with support)	≥ 0.70	0.675	fail
No-alert specificity	≥ 0.90	0.925	pass
Repeat agreement	$= 1.0$	1.0	pass
Deterministic invariance, review packet, nothing committed	required	all true	pass
Full eight-label coverage	required	false (3 labels with 0 positives)	fail

Outcome: `rejected_for_product_integration`; human review pending

Table 14: Holdout per-label results with denominators (evidence class B). Undefined metrics are reported as n/a, never as 0 or 1. Also: exact match 0.773; Hamming loss 0.053; 213 false negatives and 291 false positives in total; 7 of 13 contrast groups fully correct (member accuracy 0.863).

Label (Task 7B)	Pos.	TP	FP	FN	Precision	Recall	F1	Specificity
Crisis / self-harm	298	298	66	0	0.819	1.000	0.900	0.926
Continuing threat	132	14	51	118	0.215	0.106	0.142	0.952
Legal urgency	209	152	0	57	1.000	0.727	0.842	1.000
Coercion (comm. safety)	216	216	162	0	0.571	1.000	0.727	0.834
Explicit human request	118	80	12	38	0.870	0.678	0.762	0.989
Immediate danger	0	0	0	0	n/a	n/a	n/a	1.000
Isolation / boycott / displ.	0	0	0	0	n/a	n/a	n/a	1.000
Medical urgency	0	0	0	0	n/a	n/a	n/a	1.000
Macro, 5 defined labels (specificity over all 8)					0.695	0.702	0.675	0.963
Micro					0.723	0.781	0.751	—

6.4 Task 7B on the exposed regression fixtures (C)

Task 7B is *worse* than Task 7 on the exposed fixtures (Table 12, Figure 6). Seven-label micro F1 fell from 0.673 to 0.564 on dev and from 0.683 to 0.622 on candidates. Crisis recall is 1.00 on both splits (7/7 and 9/9), but crisis precision is only 0.41 and 0.56 (10 and 7 false positives). On the red-team victim-input cases it matched 7 of 8; the Hinglish crisis case RT-VI-HG-001 is still missed.

6.5 Deterministic invariance and reproducibility

Invariance is the property that matters most for safety: whatever the shadow model does must not change the authoritative pipeline. Sixty records were each run under seven shadow conditions: the real model, always fires, always silent, raises an error, times out, returns malformed output, and missing. The deterministic outputs were identical in every case (0 mismatches). Repeat agreement of the shadow model was 1.0, and every Stage C seed reproduced identical predictions on re-run.

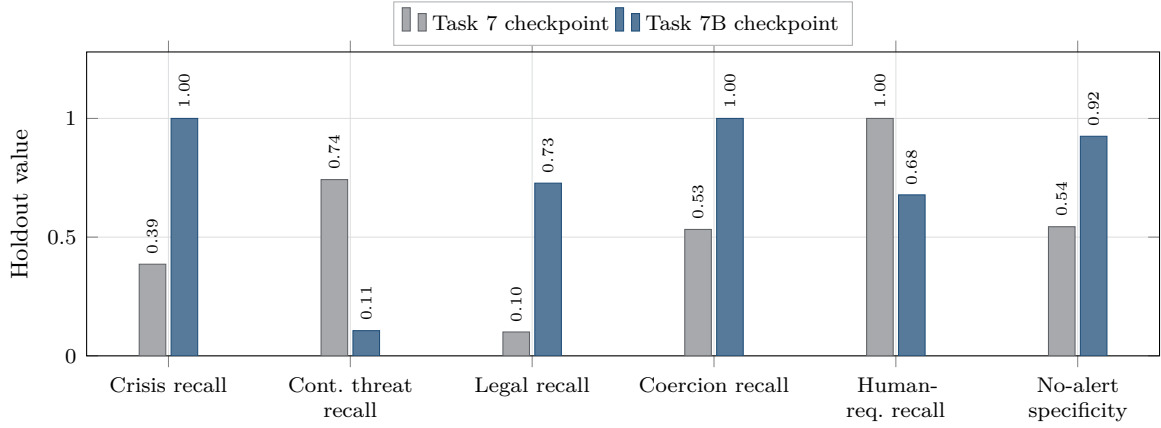


Figure 5: Same holdout, two checkpoints. Task 7B fixes crisis (0.39 $\rightarrow$ 1.00), legal (0.10 $\rightarrow$ 0.73) and coercion (0.53 $\rightarrow$ 1.00) recall and no-alert specificity (0.54 $\rightarrow$ 0.92). Continuing-threat recall collapses from 0.74 to 0.11, and human-request recall falls from 1.00 to 0.68. On this holdout Task 7’s crisis recall was 0.51 in English, 0.38 in Hindi and 0.28 in Hinglish.

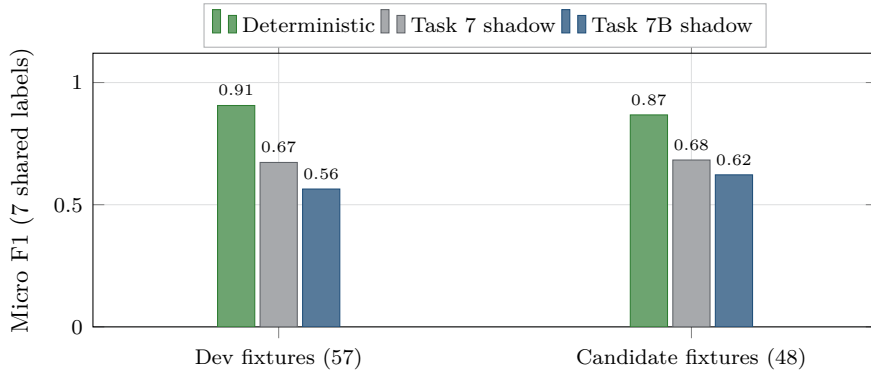


Figure 6: Contaminated exposed regression (evidence class **C**). Task 7B improves on the synthetic holdout and gets worse on the human-written fixtures: it fits the generator, not the task.

6.6 The authoritative deterministic pipeline

The deterministic text pipeline has its own evaluation. It uses no model, no network and no LLM (Table 15). The baseline (2026-09-11) used crisis lexicon **crisis-v1.1** and validator **guardrails-v1**. The hardening round moved to **crisis-v1.2**, **guardrails-v1.1** and 87 output rules. The SVI scoring version **svi-2026.09-pc08** was unchanged.

Table 15: Deterministic detectors at baseline (precision / recall / F1). Explicit human request is excluded (button, not text).

Detector	Dev (57; tuned on)			Candidate (48; exposed)		
	P	R	F1	P	R	F1
Crisis / self-harm	0.636	1.000	0.778	0.714	0.556	0.625
Immediate danger	1.000	0.875	0.933	1.000	1.000	1.000
Continuing threat	1.000	0.800	0.889	1.000	0.917	0.957
Medical urgency	1.000	1.000	1.000	1.000	0.667	0.800
Isolation / boycott / displ.	1.000	0.800	0.889	1.000	0.500	0.667
Legal urgency	1.000	1.000	1.000	1.000	0.857	0.923
Coercion (comm. safety)	1.000	0.857	0.923	1.000	0.400	0.571

Remaining failures:

- two critical misses, **DEV-EN-022** and **CAND-EN-003**;
- four false escalations, **DEV-EN-011**, **DEV-HI-012**, **CAND-EN-012** and **CAND-HI-009**, all from crisis wording that is quoted or attributed to someone else;

Table 16: Headline safety metrics, baseline vs. after hardening. Candidate numbers after hardening are regression numbers, because the failures were read while designing the fixes.

Metric	Baseline		After hardening	
	Dev	Candidate	Dev	Candidate
Critical misses / events	1/17	4/14	1/17	1/14
Wilson 95% CI of miss rate	0.010–0.270	0.117–0.547	0.010–0.270	0.013–0.315
False escalations	2	2	2	2
Crisis pre-check P / R	0.636 / 1.000	0.714 / 0.556	0.636 / 1.000	0.800 / 0.889
Evidence validity (cited turns real)	1.000	1.000	1.000	1.000
Guardrail red-team (39 cases)	20/39 pass		39/39 pass (exposed)	
Near-miss / leakage development cases	—		75/75 and 29/29 pass	

- detector false negatives concentrated in coercion, isolation and continuing threat.

The MVP target M1 (zero critical misses on the scripted corpus) is **not met**. The SVI sensitivity analysis found four things:

- without a hard override, the highest text-only SVI is 71.8 (High);
- D2 is binary in text, so its 0.18 weight never moves a band on its own;
- seven weak tier-1 signals can reach 50.91 (Moderate) with no strong indicator;
- repeated future threats stay at or below 69, under the override threshold of 70.

The reproducibility checks all passed: two runs were identical, the 10 multi-turn replays were consistent, the LLM-off walk validated all 16 fallbacks, and no network attempt was made.

External exploratory firing rates. The deterministic pipeline was also run on private samples of the external datasets. These rates are *not* correctness measures, because no SAHAY label exists for these records:

- **Reddit Suicide Detection:** the crisis pre-check fired on 53.9% of records from the suicide subreddit and 1.6% of the others.
- **Dreaddit:** 3.0% on stressed records vs. 1.7% on unstressed ones.
- **EmoInHindi:** the legal-urgency detector fired on 51.6% of records and medical urgency on 23.8%. That fits the corpus’s legal and mental-health counselling domain; it is not evidence of overbreadth.
- **Hinglish hate speech and sentiment:** essentially nothing fired (0.0–0.1%).

6.7 Model runtime qualification (Task 6)

Task 6 measured latency and memory only, on fictional text and synthetic signals (Table 17). None of these numbers is a quality measurement. Real Hindi and English WER is **unmeasured** until approved, transcribed evaluation audio exists.

The worst-case row is why Silero VAD is an *enforced* precondition rather than an optimisation. `WhisperASR.transcribe` has no default for its speech intervals, validates them before decoding, and skips Whisper when no speech is found. A VAD false positive can still reach Whisper, and that remains a documented risk.

Table 17: Runtime on the RTX 4060 laptop (fp16 CUDA unless stated). p50 over 20 runs (encoders, VAD) or 5 runs (ASR).

Component	Measurement
MuRIL (primary encoder)	Cold load 7.8 s. Encode p50 at 128 tokens: 17.8 ms (en), 14.8 ms (hi), 16.3 ms (Hinglish); 327–397 samples/s at batch 8. Tokens per character: 0.209 / 0.246 / 0.259; unknown-token rate 0. Peak 504 MB.
XLM-R (comparison only)	Cold load 4.2 s. Encode p50 at 128 tokens: 15.9 / 19.0 / 15.9 ms. Tokens per character: 0.232 / 0.319 / 0.294.
Whisper Small (faster-whisper, CTranslate2)	Cold load 1.1 s. On a speech-like <i>synthetic</i> signal: 5 s English 523 ms (real-time factor 0.105); 5 s Hindi 2,349 ms (RTF 0.47); 30 s Hindi 2,365 ms (RTF 0.08). Hindi decoding takes about 2.35 s whatever the clip length, which suggests the decoder runs to a length limit on non-speech.
Silero VAD (CPU)	p50 98 ms for 5 s of audio, 595 ms for 30 s, 1,186 ms for 60 s (about 20 ms per audio second).
Gated pipeline	10 s of silence: 0 speech intervals, Whisper never invoked, 225 ms in total.
Ungated worst case (benchmark only)	Forced to decode 5 s of <i>silence</i> in Hindi, Whisper produced a median 186 characters of fluent text; a pure tone produced 148 characters. English: 3 and 0.
Coexistence	MuRIL and Whisper together use 1,102 MB of device memory, leaving 5,950 MB of headroom on 8 GB.

7 Discussion

1. Domain adaptation worked as language modelling; its downstream value is unproven.

Stage A cut held-out perplexity by 48%, and the Stage B heads show the adapted encoder separates several source tasks. But no ablation was run against un-adapted MuRIL, XLM-R, or the planned LogReg and XGBoost baselines. The Stage C results therefore cannot be attributed to Stage A.

2. The synthetic-to-real gap dominates. On the *same* frozen holdout, Task 7B raised micro F1 from 0.50 to 0.75. On the human-written exposed fixtures it *lost* ground: 7-label micro F1 fell from 0.673 to 0.564 on dev. The most plausible reading is that the head learns the style of the template generator. A holdout drawn from the same generator family is not a proxy for real performance, however carefully it is frozen.

3. Crisis recall was bought with continuing-threat recall. Continuing-threat positives fell from 1,667 in the Task 7 training set to 453 in Task 7B’s. Task 7B’s selection rule ranks crisis recall first and, through the “weak-label” minimum, only legal and coercion recall after it. Continuing threat enters only through macro F1. The result is a recall collapse from 0.74 to 0.11 on that label (Wilson 95% CI 0.064–0.170). Crisis selection was also volatile: the chosen epoch sits on a spike (Figure 4).

4. Split design matters as much as the model. Splitting by family put *all* positives for immediate danger, isolation and medical urgency into training, leaving validation and holdout with none. The metric-validity correction made this visible: it reports those labels as undefined rather than zero-filling them and flatters nothing. Any future corpus must stratify family assignment by label.

5. The deterministic pipeline is precise but brittle. Its precision is 1.0 on almost every detector. Its failures are recall gaps on paraphrase, coercion and isolation wording, plus false escalations from quoted crisis language. A learned model could help as a *recall-oriented second opinion* for an officer. Task 7B’s crisis recall of 1.00 on both exposed splits hints at this, but its precision (0.41–0.56) and the persistent Hinglish miss (RT-VI-HG-001) show it is not ready even for that role.

6. Relation to the literature. Refs. [1, 2] report 89–97.5% accuracy for binary PTSD or emotion tasks on small or acted datasets. SAHAY-AI’s task is different: multi-label, multilingual, code-mixed,

and reported under strict evidence separation. It gives far more modest numbers, and says so. The weak severity signal in [1] ($R^2 = 0.10$) supports the decision not to derive severity from voice. MuRIL’s transliteration advantage [5] is reflected in our tokeniser measurements, yet code-mixed Hinglish remains the hardest input: it has the lowest Task 7 validation macro F1 (0.50) and the lowest Task 7 holdout crisis recall (0.28). EmoInHindi [8] is domain-relevant, but its emotion labels, and the Stage B head’s failure on *fear*, confirm that emotion is not a vulnerability proxy.

7. Governance held. Several practices worked as designed:

- the plan was hashed before training, and the corpus frozen;
- the holdout was evaluated only once;
- metrics carry denominators, and undefined values stay undefined;
- evidence classes were never pooled;
- the source-label firewall held, and shadow isolation was verified by the invariance test.

Together they turned a disappointing model into a trustworthy negative result, the outcome the project’s safety invariants are built to allow.

8 Limitations and threats to validity

- **No official or independent evaluation.** The locked set has 0 samples, and no blind corpus has been authored or frozen. Every number is class **B** or **C**.
- **Synthetic supervision.** The SAHAY labels come from agent-written templates. None of the 2,266 review-packet items has been reviewed by a person.
- **Small fixture sets.** With 17 and 14 critical events, a single miss moves the miss rate by 6–7 points. The Wilson intervals in Table 16 span 0.01–0.27 and 0.01–0.32.
- **Uncalibrated threshold.** The fixed 0.5 threshold is a convention, not an operating point. No calibration, reliability diagrams or AUPRC were computed, though the handover plan lists them.
- **Unresolved data rights.** Licensing and privacy approval for the Stage A datasets are unresolved. They contain real user-generated text, may include minors, and names are not reliably redacted. If a review refuses a dataset, the derived checkpoints must be quarantined.
- **Voice is unevaluated.** ASR quality (WER) is unmeasured, D4 is unavailable, and Whisper hallucination is mitigated but not eliminated.
- **Single machine and seed sensitivity.** Runtime numbers come from one laptop GPU. Stage C results vary by 0.07 or more in macro F1 across seeds.
- **Literature numbers were not reproduced.** The results of [1, 2] are quoted from their abstracts, not replicated.

9 Recommendations and next steps

1. **Build the independent evaluation first.** Recruit human authors and reviewers through ml/eval/blind/, freeze a blind corpus, and fill the locked set, with two reviewers on every crisis and immediate-danger sample. Until then, no model claim can move beyond development evidence.
2. **Complete the human review packet** (2,266 items) before any promotion decision.
3. **Fix corpus coverage.** Stratify family splits so every label has positive and negative support in validation and holdout. Restore continuing-threat support. Treat the 7b-v1 holdout as exposed and version a new one.
4. **Add the missing controls:** un-adapted MuRIL, XLM-R, and the planned LogReg and XGBoost baselines, all under the same protocol.

5. **Stabilise selection.** Select on a smoothed or plateau criterion, not a single-epoch spike. Report per-label AUPRC and calibration; choose any threshold on validation only.
6. **Resolve data rights** for the Stage A datasets, or retrain Stage A without the unresolved ones, and record the decision.
7. **Measure ASR** on approved, transcribed Hindi and English audio (EXT-002). Benchmark IndicConformer [6] only after its own approval.
8. **Prepare for DPDP commencement** (consent-manager provisions around November 2026, substantive obligations around May 2027), with a legal review of notices, retention and breach handling [4].

10 Conclusion

The references establish that speech and text carry trauma-relevant signal [1, 2], that NHAA 14566 needs multilingual, round-the-clock, docketed intake [3], and that India’s data-protection regime is arriving in phases [4]. They also point to the right open tools: MuRIL, IndicConformer, faster-whisper and EmoInHindi [5, 6, 7, 8].

The team’s trained model, a MuRIL-based eight-label shadow classifier, shows that domain adaptation works as language modelling. It also shows that targeted synthetic data can drive crisis recall to 1.00 on a synthetic holdout. But it does not generalise to human-written fixtures, it trades one safety label for another, and it cannot be fully evaluated on its own holdout. It therefore remains `rejected_for_product_integration`.

The deterministic pipeline stays the authority, with two known critical misses on the exposed fixtures and an honest record of every failure. The next milestone is not a better model but an independent, human-authored evaluation set. Without one, no model in this project can earn trust.

A Reproducibility and provenance

Table 18: Pinned artefacts and hashes (weights, corpora and reports stay private beneath SAHAY_TRAINING_ROOT; never committed).

Artefact	Identifier
Base encoder	<code>google/muril-base-cased @ afd9f36c7923d54e97903922ff1b260d091d202f</code> (Apache-2.0)
ASR	<code>openai/whisper-small @ 973afd24...</code> , converted locally to CTranslate2 fp16; faster-whisper 1.2.1, CTranslate2 4.8.2
VAD	<code>silero-vad 6.2.1</code>
Stage A run	<code>run-20260912-054730; model.safetensors sha256 550fc3a2...a35c; corpus external-ext119-v1, segments sha256 37f5bf5d...f844</code>
Task 7 selected checkpoint	<code>stage-c/run2-seed-97; fictional corpus sha256 f07cfbfa...97e3; template bank 9cb89828...78b0</code>
Task 7B plan	<code>sha256 843f3e2c62545e79...892b</code> , recorded 2026-09-12 11:06:48 IST
Task 7B corpus 7b-v1	frozen 2026-09-12 11:01:49 IST; template bank <code>43ac65f9...bb43</code> ; holdout <code>sha256 3fad8607...eb22</code>
Task 7B selected checkpoint	<code>task7b/checkpoints/C-seed-13, epoch 3</code>
Software	Python 3.12, torch 2.11.0+cu128, transformers 5.17.0, tokenizers 0.23.2, accelerate 1.15.0, safetensors 0.8.0, numpy 2.5.3

Commands (private model environment, offline with sockets blocked, explicit SAHAY_DATASETS_ROOT, SAHAY_MODELS_ROOT and SAHAY_TRAINING_ROOT):

```
python -m ml.data.training_corpus build -ext119-local-training -acknowledge "<EXT-119 sentence>"
python -m ml.training.cli preflight | fictional | stage-a | stage-b
python -m ml.training.cli stage-c -seeds 13 42 97 -epochs 20 -patience 3 -tag run2
python -m ml.training.cli verify-shadow | regression | error-analysis
python -m ml.training.cli hardening-build | hardening-verify | stage-c7b-plan
python -m ml.training.cli stage-c7b -phase 1 | stage-c7b -phase 2 | stage-c7b-select
python -m ml.training.cli stage-c7b-evaluate | stage-c7b-correct
python -m ml.eval.run_eval # deterministic evaluation (standard library only)
python -m ml.eval.hardening_report
```

Project-internal sources used for this report.

- Repository: ml/training/README.md, ml/shadow/MODEL_CARD.md, ml/eval/README.md, ml/runtime/README.md, ml/data/cards/emoinhindi.md.
- Baseline evaluation: ml/eval/results/eval-baseline-2026-09-11.md.
- Hardening report: ml/eval/results/safety-hardening-2026-09-11.md.
- Specification and decisions: docs/HANDOVER.md and docs/EXTERNAL_DECISIONS.md (EXT-118 to EXT-121).
- Private reports under sih-dataset/training/ and sih-dataset/reports/. They hold aggregate JSON only; no record text was read for this report.

B Label definitions

Table 19: The eight detector categories (label schema 1.0.0) and the deterministic signal each is judged on.

Category	Code identifier	Deterministic signal
Crisis / self-harm	<code>crisis_self_harm</code>	The synchronous crisis pre-check fires on the turn
Immediate danger	<code>immediate_danger</code>	A D1 tier-3 (imminent presence) lexicon match
Continuing threat	<code>continuing_threat</code>	Any D1 match, or a D3 match of tier 2 or above
Medical urgency	<code>medical_urgency</code>	A D7 match of tier 2 or above (pain alone is tier 1)
Isolation / boycott / displacement	<code>isolation_boycott_displacement</code>	Any D6 match
Legal urgency	<code>legal_urgency</code>	Any D8 match
Coercion (communication safety)	<code>communication_safety_coercion</code>	Any D9 match, or evidence behind the coercion alert
Explicit human request	<code>explicit_human_request</code>	Excluded: the app sends <code>request_human</code> from a button

A *critical miss* is a sample whose expected routing is Critical but that was neither caught by the pre-check nor banded Critical. A *false escalation* is a sample routed Critical when it should not have been. Labels describe what the text *means*, not what the system does. A crisis phrase quoted in someone else’s words is labelled `crisis_self_harm = false`.

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All URLs accessed 13 September 2026. PIB and MeitY pages refused automated access (HTTP 403); their content was verified through the PIB mirror and public summaries of the gazette notification.